

Education

B.Tech, Electrical Engineering , Indian Institute of Technology Goa	CGPA: 7.23/10	2023 – Present
Class XII, CBSE , Rishikul Public Sr. Sec. School, Jind	Aggregate: 91.8%	2021 – 2022

Experience

STEP COUNTER using VHDL, Arduino, ADXL345 and DE10-Lite FPGA

- Designed and implemented a digital step counter.
- Interfaced ADXL345 accelerometer with Arduino.
- Displayed and verified output using DE10-Lite FPGA board.

Calculator using VHDL

- Developed a calculator using VHDL.
- Simulated and tested the design on DE10-Lite FPGA.

Puzzle Game

- Developed a puzzle game in C++.
- Built a web version using HTML, CSS and JavaScript.

Gym Supplement Store Website

- Designed and developed an e-commerce style website.
- Technologies used: HTML, CSS and JavaScript.
- Website: [Netlify](#)

Three Phase Fault Analysis using MATLAB Simulink

- Simulated three-phase fault analysis.
- Implemented overcurrent relay protection.

Smart Door Lock

- Developed an IoT based smart door lock.
- Used ESP32, Camera, Keypad, Bluetooth and Wi-Fi.

Electrical Calculator

- Developed using C++, VS Code.

PID Controller

- Designed and simulated a PID Controller using MATLAB Simulink.

Skills

Programming	C++, Python, HTML, CSS, JavaScript
Electrical Tools	MATLAB, Simulink, AutoCAD, FEM Magnet Software, MS Word, MS Excel
Hardware	VHDL, Arduino UNO, STM32, ESP32, DE10-Lite FPGA
Relevant Coursework	Power Electronics, Power Systems, IoT, Digital Integrated Circuit Design, Control Systems, Digital Communication, Digital Signal Processing, Electromagnetic Waves, Electrical Machines, Probability and Random Processes, Analog Communication, Analog Circuits, Signals and Systems, Electronic Devices, Digital Circuits
Workshops	PCB Designing, Arduino

Positions of Responsibility

Student Design Editor	Designed templates and contributed to the layout of Gurukul Varta (IIT Goa Newsletter).
Event Organizer	Contributed to organizing Cultural Fest, Independence Day, Republic Day and various institute-level events.

Extracurriculars & Hobbies

- Playing Cricket and Badminton
- Travelling and Listening to Music
- Hosting and organizing institute events at IIT Goa